

**MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL**

**MID-TERM Exam (Sept. 2024)**

**Department of Electrical Engineering**

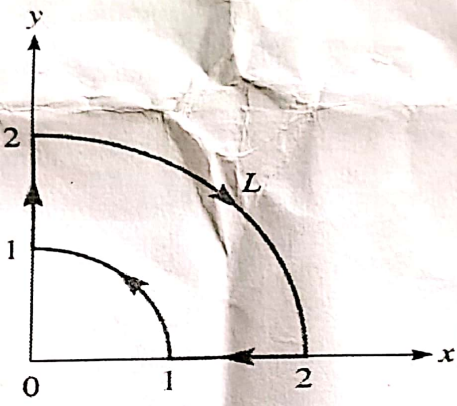
**B.Tech III Sem (Electrical)**

**Sub: Electromagnetic Fields and Materials (EE-211)**

**Time: 1.5 hr.**

**Max. Marks: 20**

**NB: Assume suitable data if necessary**

| Q. No. | Question                                                                                                                                                                                                                                                                                                                                                                      | Marks    |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1      | An electric field intensity is given as $= \left( \frac{100 \cos \theta}{r^3} \right) \hat{r} + \left( \frac{50 \sin \theta}{r^3} \right) \hat{\theta}$ . At the point whose spherical coordinates are $(r = 2, \theta = 60^\circ, \phi = 20^\circ)$ . Find: (a) Magnitude of electric field (b) a unit vector (in Cartesian coordinates) in the direction of electric field. | 04       |
| 2      | Let $A = \rho \sin \phi a_\rho + \rho^2 a_\phi$ . Evaluate $\oint_L A \cdot dl$ if L is the contour surface given in the figure.<br><br>                                                                                                                                                    | 04       |
| 3      | Derive the expression for the Electric field due to a infinite surface charge                                                                                                                                                                                                                                                                                                 | 04       |
| 4      | Let $D = 2xya_x + x^2a_y$ C/m <sup>2</sup> .<br><div style="text-align: right; margin-right: 100px;"><math>0 \rightarrow z</math></div> a) The volume charge density $\rho_v$ .<br><div style="text-align: right; margin-right: 100px;"><math>0 &lt; y &lt; 1</math></div> b) The total charge contained in the region $0 < x < 1, 0 < z < 1, y = 1$                          | 02<br>02 |
| 5      | An electric field is expressed by $E = 6x^2\hat{x} + 6y\hat{y} + 4z\hat{z}$ V/m. Find<br>(a) $V_{MN}$ if points M and N are specified by M(2,6,-1) and N (-3, -3, 2)<br>(b) $V_M$ if $V=0$ at Q (4, -2, -35)<br>(c) $V_N$ if $V=2$ at P (1, 2, -4)                                                                                                                            | 04       |

-----End-----